DGMS 2019 • SECTION K: ENGINEERING GEOLOGY & TUNNELING

Question 85: Tunnel Site Selection & Geological Conditions: Horizontal Strata

Official Examination Guide • Structural Controls in Underground Excavations: Folds vs. Flat-Lying Formations

OFFICIAL EXAMINATION QUESTION

Which one of the following is the best condition for tunnel site consideration?

(a) Tunnel along the axis of an anticline

(b) Tunnel within thick horizontal formation CORRECT ANSWER

(c) Tunnel along the axis of a syncline

(d) Horizontal tunnel parallel to the dip direction of rock strata

1. Executive Summary & Core Engineering Principle

In underground tunneling and rock engineering (as detailed in **K.M. Bangar**, *Principles of Engineering Geology*, Chapter 11, pp. 370–374; **P.K. Mukherjee**; and **F.G. Bell**, *Engineering Geology*), structural discontinuities and fold

geometry fundamentally govern roof stability, rock pressure, and groundwater inflow:

- **(b) Tunnel within Thick Horizontal Formation (BEST CONDITION):** When a tunnel is driven inside a thick, massive, flat-lying formation:
    1. *Lithological Uniformity:* The excavation advances entirely through a single rock type of uniform strength and elastic properties, eliminating unexpected lithological contacts and differential overbreak.
    2. *Beam / Bridge Action:* Thick horizontal bedding acts as a natural structural beam across the tunnel crown, providing self-supporting arch action and minimizing roof falls.
    3. *Symmetric In-situ Stress:* The vertical overburden stress ($\sigma_v = \rho g z$) acts perpendicularly to the bedding planes, creating symmetric stresses on side walls with zero asymmetric shearing or sliding forces.
    4. *Benign Groundwater Regimes:* Avoids artesian folded troughs; groundwater drains predictably without building dangerous hydraulic heads.
- **(a) Tunnel along the Axis of an Anticline (Unfavorable):** The outer arc of an anticlinal crest undergoes intense **tensile stretching** during folding, developing extensive open extensional joints and fractures. The rock mass at the crest is heavily shattered, leading to severe roof caving, rock bursts, and enormous overbreak.
- **(c) Tunnel along the Axis of a Syncline (Worst Possible Condition):** A syncline acts as a giant natural hydrological funnel. Permeable beds in both limbs channel groundwater directly into the synclinal trough under high **artesian hydrostatic pressure**, causing catastrophic flooding, heavy pumping demands, and extreme hydrostatic loads on tunnel linings.
- **(d) Tunnel Parallel to the Dip Direction (Unfavorable):** Driving parallel to the dip direction forces the tunnel to intersect constantly changing rock layers of varying hardness, permeability, and strength, resulting in non-uniform overbreak and sliding of inclined beds into the tunnel profile.

Therefore, the **best condition** for tunnel site consideration is **(b) Tunnel within thick horizontal formation**.

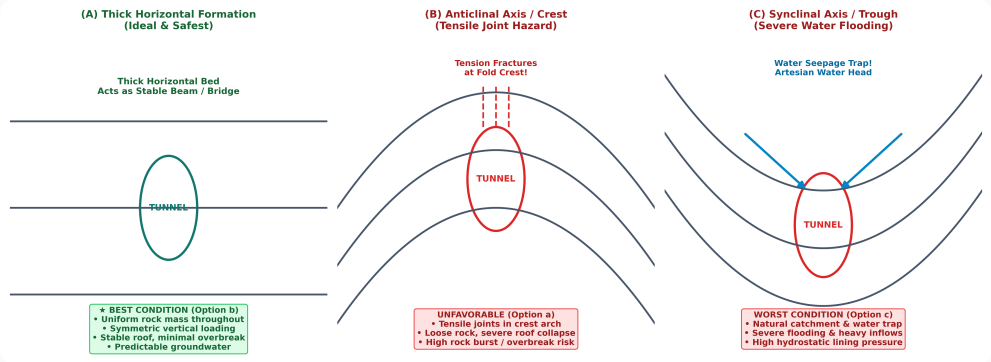


Figure 85.1: Comparative stability and groundwater conditions in tunneling through Thick Horizontal Strata (A, Ideal/Best), Anticlinal Crest (B, Tensile Fractures), and Synclinal Trough (C, Artesian Flooding Hazard) (Grounded in Bangar Engineering Geology and Bell).

2. Comparative Engineering Evaluation of Tunneling Geological Settings

Geological Setting	Suitability	Roof & Crown Stability	Groundwater Inflow Behavior	Overbreak & Support Requirements
Thick Horizontal Formation	**BEST / IDEAL**	Excellent; massive horizontal beds act as stable self-supporting beams across the roof span.	Predictable, gravity-drained; no artesian hydrostatic pressure traps.	Minimal overbreak; standard uniform rock bolt and shotcrete pattern.
Anticlinal Crest / Axis	**Unfavorable**	Poor; extensional tension joints cause loose, blocky rock and severe roof caving.	Generally dry (sheds water into limbs), but gas accumulation may occur.	Excessive overbreak; heavy steel arches and forepoling required.

Geological Setting	Suitability	Roof & Crown Stability	Groundwater Inflow Behavior	Overbreak & Support Requirements
Synclinal Trough / Axis	**WORST HAZARD**	Compression in the trough causes crushing, spalling, and heavy swelling pressures.	**Catastrophic**; natural water catchment under high artesian head causing severe flooding.	Massive support and expensive impervious pressure linings required.
Across Strike (Perpendicular to Dip)	**Good**	Shortest exposure length per individual bed; bridge action across dip.	Manageable seepage.	Moderate support needed; much better than tunneling along dip direction.
Parallel to Dip Direction	**Poor**	Daylighting beds slide into the tunnel face; asymmetric wall pressures.	Seepage along dipping bedding planes enters tunnel asymmetrically.	Frequent variations in support design as each stratum is crossed.

3. Systematic Option Evaluation

- **(a) Tunnel along the axis of an anticline: INCORRECT.** Tension cracks at the anticlinal crest make the roof severely fractured and unstable.
- **(b) Tunnel within thick horizontal formation: CORRECT ANSWER.** Provides uniform lithology, symmetric stress distribution, natural roof beam stability, and benign hydrological conditions.
- **(c) Tunnel along the axis of a syncline: INCORRECT.** Synclinal troughs act as natural water traps under high artesian pressure, presenting the worst flooding hazards in engineering geology.

- **(d) Horizontal tunnel parallel to the dip direction of rock strata: INCORRECT.** Causes continuous changing of lithology, asymmetrical side-wall thrusts, and sliding along dipping bedding planes.

> **Key Takeaway:** Thick horizontal formation = Ideal self-supporting roof beam + uniform rock; Anticlinal axis = Tension fractures and roof falls; Synclinal axis = Artesian water flooding hazard. Hence, **(b) Tunnel within thick horizontal formation** is the best condition.

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